

# Unit Economics Framework: The \$20/Tonne Logistics Limit

## Executive Summary

By integrating our geochemical dissolution limits with real-world field deployment economics, we have derived the exact break-even constraints for any ERW operation. The single most important variable is not the rock chemistry, not the soil type, and not the carbon price — it is the logistics cost per tonne of delivered basalt.

## The Carbon Math

### Theoretical Maximum

- Basalt has a theoretical maximum CDR potential of ~0.3 tCO<sub>2</sub> per tonne of rock upon complete dissolution
- At an application rate of 25 t/ha: theoretical maximum = 7.5 tCO<sub>2</sub>/ha

### Empirical Reality (Our Data)

- Maximum observed dissolution: ~24.7% (capped by passivation layer)
- Realistic CDR yield at 25 t/ha: ~1.85 tCO<sub>2</sub>/ha
- Only Year 1 carbon is bankable; Year 2+ reverts to background

### Revenue Per Hectare

| Carbon Price           | CDR Yield (tCO <sub>2</sub> /ha) | Revenue/ha |
|------------------------|----------------------------------|------------|
| \$150/tCO <sub>2</sub> | 1.85                             | \$278      |
| \$200/tCO <sub>2</sub> | 1.85                             | \$370      |
| \$300/tCO <sub>2</sub> | 1.85                             | \$555      |

# The Cost Structure

## Fixed Costs (Per Tonne of Rock)

| Cost Component               | Range          | Notes                                                            |
|------------------------------|----------------|------------------------------------------------------------------|
| Rock procurement (ex-quarry) | 2–8/t          | Crusher dust / quarry waste is cheapest                          |
| Crushing & sizing            | 3–12/t         | Ultra-fine grinding is 3–4× more expensive and counterproductive |
| Transportation               | 5–25/t         | THE critical variable — function of distance                     |
| Spreading & application      | 2–5/t          | Standard agricultural equipment                                  |
| MRV & monitoring             | 1–3/t          | Our stack reduces this by ~80% vs lab-heavy approaches           |
| <b>Total delivered cost</b>  | <b>13–53/t</b> |                                                                  |

## The Break-Even Analysis

At a carbon price of \$150/tCO<sub>2</sub> and our observed 24.7% dissolution:

- Revenue per tonne of rock: ~\$11.10
- **Maximum allowable total cost per tonne: \$11.10**
- At 200/tCO<sub>2</sub> : *Maximum allowable cost is to 14.80/t*
- At 300/tCO<sub>2</sub> : *Maximum allowable cost is to 22.20/t*

## The Logistics Kill Zone

Because rock procurement (2–8) and crushing (3–12) are relatively fixed, the only variable that determines profitability is **transportation distance**.

## Rule of Thumb

- Truck freight in India: ~₹35/km for a 5-tonne truck
- At 20 km distance: ~\$4/t transport cost → **Profitable**

- At 50 km distance: ~\$10/t transport cost → **Marginal**
- At 100 km distance: ~\$20/t transport cost → **Unprofitable at most carbon prices**

## The Agronomic Subsidy

The yield gains we documented (207–443 per farmer per season) create an implicit subsidy:

- Farmers accept the rock for free because it acts as a premium soil amendment
- No customer acquisition cost
- No land lease or access fees
- The farmer becomes a willing distribution partner

This is the critical insight most competitors miss: the agronomic value of ERW is the business model's foundation, not the carbon credits. Carbon credits are the profit margin on top of a self-sustaining agricultural value chain.

## What We Are Not Sharing Here

- Our proprietary logistics optimization model (quarry-to-farm routing)
- The specific quarry sources and their delivered cost structures
- Our geographic targeting maps showing optimal deployment zones
- The exact financial model linking dissolution kinetics to credit issuance timing

# Economic Cost Structure

